Edge-Based Predictive Maintenance for Embedded Devices

Research Project Proposal • Md Mamun Hossain • Computer Science & Engineering

Proposed by	Md Mamun Hossain
Project type	Research / prototype study
Estimated duration	8–10 weeks
Max collaborators	3

Abstract

I want to investigate how lightweight machine-learning models can be used to identify early signs of abnormal behavior in embedded devices. The project will analyze sensor and system-health signals such as temperature, vibration, voltage, and operating patterns and investigate whether a compact model can detect unusual behavior before a potential hardware failure.

Research Motivation

Embedded devices are increasingly used in monitoring, automation, and connected systems. A small hardware problem can become costly when it is discovered only after a device stops working. I want to explore whether edge-based analysis can provide an earlier warning while keeping computation close to the device and reducing dependence on continuous cloud processing.

Research direction: The project will start with a small and measurable prototype. I will compare a simple statistical or rule-based baseline with a lightweight machine-learning approach and study the trade-off between detection quality, latency, and resource usage.

Research Objectives

- Collect or simulate sensor and system-health measurements from an embedded-device scenario.
- Identify useful features for detecting abnormal device behavior.
- Develop a lightweight anomaly-detection or classification model suitable for edge deployment.
- Compare model performance with a simple baseline approach.
- Measure detection quality, inference time, and resource requirements.

Research Questions

RQ1. Which sensor or system-health features provide useful early signals of abnormal behavior?

RQ2. Can a lightweight model detect abnormal conditions with acceptable accuracy?

RQ3. What trade-off exists between model complexity and edge-device resource usage?

RQ4. How early can an abnormal pattern be detected before a simulated failure condition?

Proposed Methodology

Step 1. Review research on predictive maintenance, anomaly detection, and edge AI for embedded systems.

Step 2. Define a small embedded-device scenario and select relevant sensor/system-health variables.

Step 3. Collect, generate, or use an appropriately licensed dataset containing normal and abnormal operating patterns.

Step 4. Clean the time-series measurements and create meaningful statistical or temporal features.

Step 5. Establish a simple threshold or statistical baseline.

Step 6. Train and evaluate a lightweight machine-learning model.

Step 7. Measure inference latency, memory/CPU usage, and detection performance on an appropriate local or edge environment.

Step 8. Analyze false alarms, missed detections, and cases where the model cannot confidently classify the device state.

Data and Experimental Resources

The project can begin with a public or simulated sensor dataset and later be extended to a small microcontroller-based setup if hardware is available. Candidate measurements include temperature, vibration, voltage, current, or other non-sensitive device-health signals. The experiment will remain small enough to reproduce with student-level resources.

Evaluation Plan

The model will be evaluated using suitable classification or anomaly-detection metrics such as precision, recall, F1-score, false-alarm rate, and detection latency. Resource usage will also be measured where possible. Results will be compared with the baseline so that any improvement can be discussed in practical terms.

Expected Challenges

- Real hardware failures are difficult to reproduce safely and consistently.
- Sensor readings can be noisy, incomplete, or affected by environmental conditions.
- A model that performs well offline may be too expensive for a small embedded device.
- A limited dataset may not represent every possible failure pattern.

Reproducibility and Documentation

I will document the dataset or simulation settings, sensor variables, preprocessing steps, feature definitions, model configuration, evaluation metrics, and edge-deployment settings. The main experiments will be organized so that a collaborator can review and reproduce the core comparison.

Planned Mitigation

I will begin with a simulated or controlled dataset and a simple baseline before moving to hardware deployment. If hardware limitations prevent full deployment, the research will focus on edge-resource measurements in a representative environment rather than claiming real-world failure prediction.

Expected Outcomes

- A documented sensor-data preparation and feature-engineering pipeline.
- A lightweight anomaly-detection or predictive-maintenance prototype.
- Comparison between a baseline method and a machine-learning approach.
- Analysis of detection performance, false alarms, latency, and resource usage.
- A research report discussing practical edge-AI limitations and future improvements.

Interest Tags

Machine Learning • Artificial Intelligence • Robotics • Embedded Systems • Edge AI • Predictive Maintenance • Industrial Automation

Required Skills

Python • Embedded Systems • Microcontrollers • Machine Learning • Deep Learning • Time-Series Analysis

Technical Stack

Python, NumPy, pandas, scikit-learn, TensorFlow Lite or PyTorch where appropriate, microcontroller/embedded hardware, and Git.

Potential Research Collaboration

I am looking for collaborators interested in embedded systems, machine learning, edge AI, or predictive maintenance. A collaborator could contribute to sensor-data preparation, feature engineering, model design, hardware integration, or evaluation.

Scope and Responsible Research

The project will use non-sensitive device-health measurements and controlled experiments. Hardware tests will be performed only on devices that are safe and authorized for experimentation. The prototype will be treated as an academic research system and will not be presented as a safety-critical industrial monitoring solution without further validation.

Proposed Work Plan

Phase	Activities
1. Background study	Literature review, problem definition, feasibility check
2. Data preparation	Sensor data selection, cleaning, labeling, feature design
3. Baseline	Implement threshold/statistical baseline
4. ML experiments	Train, compare, and optimize lightweight models
5. Edge evaluation	Measure latency and resource usage
6. Reporting	Error analysis, limitations, documentation, presentation

What I Am Looking for in a Collaborator

I would like to work with someone who is genuinely interested in the research problem and can contribute to embedded programming, data analysis, model evaluation, or experimental design. I am open to refining the scope after discussing available hardware, datasets, and time constraints.

Possible Next Step

The first step would be to agree on the embedded-device scenario, select the most appropriate dataset or sensing setup, and define a small baseline experiment before moving to model development and edge deployment.

Note: This document describes a proposed research project. It does not claim completed research, published results, or guaranteed predictive performance. The methodology and scope can be refined through collaboration.